

Positive Interpolation Rigidity, Geometric Newton Expansions, and Integer-Cell Dynamics

New progress toward the Alaoglu–Erdős conjecture

Research companion report

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Abstract

Let $x \in \mathbb{R}$ and suppose that $a = 2^x$ and $b = 3^x$ are integers. The Alaoglu–Erdős conjecture asserts that x is an integer. This document is a companion to the supplied unified report, not a synopsis of it. It deliberately avoids reproducing that report’s determinant, transcendence, p -adic, moment-matrix, and formalization programs. The principal new result is an exact four-node covariance identity: a power series with nonnegative coefficients that assumes the values $(1, a, b, ab)$ at $(1, 2, 3, 6)$ must be a single monomial. This gives a sharp rigidity theorem, a quantitative stability estimate, an explicit Farkas-type separating functional, and a weighted lower bound on the negative mass of every signed interpolant.

A second new line develops the Newton interpolation of $z^{\log_2 3}$ on the geometric grid $1, 2, 4, \dots$. Its coefficients admit a closed product formula, have an eventually alternating sign, and produce a one-signed tail at every integer between consecutive powers of 2. Rewriting the terms as a basic hypergeometric series produces a canonical entire interpolant of $2^m \mapsto 3^m$ and isolates a concrete arithmetic comparison problem. A third line packages every hypothetical counterexample into a positive integer cocycle over the irrational rotation $n \mapsto \{nx\}$; primitive counterexamples impose exact 2-adic or 3-adic signatures on that cocycle.

No complete proof is claimed. The report identifies exactly what has been proved, why several tempting completions fail, and four narrowly formulated next targets. Competitive announcements mentioned in the prompt are not used as mathematical premises; every result below is proved from stated assumptions.

Contents

1	Mandate, nonduplication, and status	3
2	Baseline reductions and a primitive normalization	3
3	The four-node positivity identity	4
3.1	Absolutely monotone interpolation	5
4	Quantitative stability and denominator quantization	6
5	The discrete moment cone and an explicit global separator	6
6	A cubic diagnostic interpolant	8

7 A two-moment Jensen rigidity theorem	8
8 The integer chord cocycle	9
9 Geometric Newton interpolation on powers of two	10
9.1 A canonical entire interpolant	11
9.2 The denominator wall	12
10 Signed interpolation as an optimization problem	12
11 Renormalized iterates and why derivative extraction is too weak	13
12 Other routes tested and their present obstruction	13
12.1 Continued fractions of α	13
12.2 Finite differences on 3-smooth nodes	14
12.3 Positive approximation without exact interpolation	14
12.4 The canonical q -interpolant	14
13 Computational reconnaissance	14
14 Four focused next targets	15
14.1 Target A: a unit-cell theorem for the q -interpolant	15
14.2 Target B: an arithmetic lower bound for weighted negativity	15
14.3 Target C: an adelic obstruction for the chord cocycle	15
14.4 Target D: a machine-checked positivity core	15
15 Lean formalization blueprint	16
16 Conclusions	16
A Proof of the general geometric divided-difference formula	17
B Reproducible numerical scan	17

1 Mandate, nonduplication, and status

The supplied source is a large cumulative research report. Its main families of ideas include interpolation determinants, transcendence conjectures, p -adic logarithms, auxiliary functions, matrix positivity, moment methods, approximation programs, and formalization plans. Repeating those programs under new names would add little. The present report therefore concentrates on three constructions that are structurally different:

- (i) a *coefficientwise positive interpolation cone* and its exact covariance defect;
- (ii) a *geometric Newton basis* whose coefficients can be evaluated in closed form;
- (iii) an *integer-valued chord cocycle* attached to the fractional parts of nx .

The elementary reductions in [Section 2](#) are included only to make the document logically self-contained.

PROVED IN THIS REPORT.

The central positivity theorem, its measure-theoretic form, quantitative stability, the global separator, the signed-negativity budget, the geometric-Newton coefficient formula, and the integer cocycle are unconditional.

OPEN BOTTLENECK.

The missing implication is not a technicality: integrality of 2^x and 3^x does not presently yield a positive integer-exponent representation. The report turns that vague gap into explicit arithmetic subproblems, but does not close it.

2 Baseline reductions and a primitive normalization

Throughout, put

$$a = 2^x, \quad b = 3^x, \quad \alpha = \log_2 3.$$

Then $b = a^\alpha$ and $6^x = ab$. Since a positive integer cannot lie strictly between 0 and 1, $x \geq 0$; the case $x = 0$ is immediate.

Proposition 2.1 (Rational and algebraic cases). *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. If x is algebraic and both 2^x and 3^x are algebraic, then $x \in \mathbb{Z}$.*

Proof. Write $x = p/q$ in lowest terms with $q > 0$. If $2^{p/q} = a \in \mathbb{Z}$, then $a^q = 2^p$; comparison of the 2-adic valuations gives $q \mid p$, hence $q = 1$. For algebraic irrational x , the Gelfond–Schneider theorem says that every value of 2^x is transcendental, contradicting $2^x \in \mathbb{Z}$. The remaining algebraic possibilities are rational and were just handled. $\square$

Thus a counterexample would necessarily have x transcendental. It would also give the 2×2 algebraic array required to contradict the four exponentials conjecture: the two logarithms $\log 2, \log 3$ are linearly independent over $\mathbb{Q}$, as are $1, x$ when $x \notin \mathbb{Q}$, while all four exponentials $2, 3, 2^x, 3^x$ would be algebraic. The six exponentials theorem does not directly apply because no third independent logarithmic direction is available; this is precisely why the problem is delicate [13, 9].

A small normalization is useful in every arithmetic approach.

Proposition 2.2 (Primitive counterexample reduction). *Suppose (x, a, b) is a nonintegral counterexample. Let*

$$r = v_2(a), \quad s = v_3(b), \quad t = \min(r, s).$$

Then $y = x - t > 0$ is again a nonintegral counterexample, with

$$2^y = \frac{a}{2^t} \in \mathbb{Z}, \quad 3^y = \frac{b}{3^t} \in \mathbb{Z},$$

and at least one of 2^y and 3^y is a unit at its distinguished prime: either $v_2(2^y) = 0$ or $v_3(3^y) = 0$.

Proof. The displayed quotients are integers by the definition of t . Since $2^t \leq a = 2^x$, one has $t \leq x$, so $y \geq 0$. If $y = 0$, then $x = t \in \mathbb{Z}$, contrary to hypothesis. Subtracting an integer does not turn a noninteger into an integer. Finally, one of $r - t$ and $s - t$ is zero. $\square$

Accordingly, any minimal or normalized counterexample may be assumed to satisfy

$$a \text{ odd} \quad \text{or} \quad 3 \nmid b. \tag{1}$$

This weak-looking condition later gives exact congruence information for the integer cocycle in Section 8.

3 The four-node positivity identity

The point of departure is to replace direct approximation of z^x by a question about the cone generated by the integer monomials $1, z, z^2, \dots$

Theorem 3.1 (Four-node covariance identity). *Let*

$$F(z) = \sum_{n \geq 0} c_n z^n,$$

where $c_n \geq 0$ and $F(6) < \infty$. Then

$$\begin{aligned} F(1)F(6) - F(2)F(3) &= \sum_{0 \leq n < m} c_n c_m (2^m - 2^n)(3^m - 3^n) \\ &\geq 0. \end{aligned} \tag{2}$$

Equality holds if and only if at most one coefficient c_n is nonzero.

Proof. All sums are absolutely convergent, so they may be rearranged. Pair the ordered terms (n, m) and (m, n) in

$$\sum_{n, m} c_n c_m (6^m - 2^n 3^m).$$

The diagonal terms vanish, while an unordered pair $n < m$ contributes

$$c_n c_m (6^m + 6^n - 2^n 3^m - 2^m 3^n) = c_n c_m (2^m - 2^n)(3^m - 3^n).$$

Every factor is nonnegative and is strictly positive for $n < m$. Hence equality forces $c_n c_m = 0$ for every distinct pair, which is equivalent to support of cardinality at most one. $\square$

Corollary 3.2 (Positive-interpolant rigidity). *Suppose $F(z) = \sum c_n z^n$ has nonnegative coefficients, $F(6) < \infty$, and*

$$F(1) = 1, \quad F(2) = 2^x, \quad F(3) = 3^x, \quad F(6) = 6^x.$$

Then $x \in \mathbb{N}$ and $F(z) = z^x$.

Proof. The target values satisfy $F(1)F(6) = F(2)F(3)$. By Theorem 3.1, $F(z) = cz^k$ for some $k \in \mathbb{N}$. The value at 1 gives $c = 1$, and the value at 2 gives $2^k = 2^x$, hence $x = k$. $\square$

The identity is not tied to the numbers 2 and 3.

Theorem 3.3 (Measure form). *Let μ be a probability measure on a totally ordered subset of $\mathbb{R}$, and let*

$$M(u) = \int u^t d\mu(t)$$

be finite at u, v, uv , where $u, v > 1$. Then

$$M(uv) - M(u)M(v) = \frac{1}{2} \iint (u^s - u^t)(v^s - v^t) d\mu(s)d\mu(t) \geq 0. \quad (3)$$

Equality holds if and only if μ is a point mass.

Proof. Expand the right-hand side and use symmetry. Because u^t and v^t are strictly increasing functions of t , the integrand is nonnegative and vanishes exactly on the diagonal $s = t$. Equality therefore implies that two independent samples from μ coincide almost surely, which is possible only for a point mass. $\square$

In probabilistic language, if N is integer-valued, then

$$M(6) - M(2)M(3) = \text{Cov}(2^N, 3^N).$$

The two random variables are strictly comonotone, so zero covariance is a degeneracy certificate. This is the shortest conceptual proof of Corollary 3.2.

3.1 Absolutely monotone interpolation

A function analytic at the origin with nonnegative Taylor coefficients is absolutely monotone on every positive interval inside its disk of convergence. Conversely, standard Bernstein-type representation theorems turn absolute monotonicity into coefficientwise positivity under mild analytic hypotheses. Thus Corollary 3.2 may be read as follows.

Corollary 3.4. *Let F be analytic on a disk centered at the origin with radius greater than 6, and suppose every Maclaurin coefficient of F is nonnegative. If F interpolates*

$$(1, 1), \quad (2, a), \quad (3, b), \quad (6, ab)$$

then $(a, b) = (2^k, 3^k)$ for an integer $k \geq 0$.

This identifies a sharply delimited bridge: any construction that derives an absolutely monotone interpolant from the integrality of a and b would finish the conjecture immediately. The later separator theorem explains why an exact construction cannot arise from a generic positivity-preserving interpolation routine; it has to use the arithmetic hypothesis in an essential way.

4 Quantitative stability and denominator quantization

The exact identity has a useful stability form. Let $(p_n)_{n \geq 0}$ be a probability distribution with $M(6) < \infty$, and put $M(t) = \sum p_n t^n$ and

$$\Delta = M(6) - M(2)M(3).$$

Proposition 4.1 (Concentration from a small defect). *One has*

$$\Delta \geq 1 - \sum_{n \geq 0} p_n^2 \geq 1 - \sup_n p_n. \quad (4)$$

Consequently, if $\Delta \leq \varepsilon < 1$, some atom has mass at least $1 - \varepsilon$.

Proof. For $m > n$,

$$(2^m - 2^n)(3^m - 3^n) = 6^n(2^{m-n} - 1)(3^{m-n} - 1) \geq 2.$$

Insert this in (2):

$$\Delta \geq 2 \sum_{n < m} p_n p_m = 1 - \sum_n p_n^2.$$

Since $p_n^2 \leq (\sup_j p_j) p_n$, the second inequality follows. $\square$

There is also an exact arithmetic quantization.

Corollary 4.2 (Rational-weight gap). *Let $D \geq 1$. If every p_n is an integral multiple of $1/D$, then $D^2 \Delta$ is a nonnegative integer. Hence either $\Delta = 0$ or $\Delta \geq D^{-2}$.*

Proof. Write $p_n = r_n/D$ with $r_n \in \mathbb{N}$. Multiplying (2) by D^2 gives a sum of nonnegative integers. $\square$

This gives a possible combinatorial completion criterion. It is enough to construct a rational positive model whose covariance defect is strictly smaller than the square of its weight mesh. The construction must be extraordinarily sharp: the target lies on the equality boundary, not in the interior of the moment cone.

Remark 4.3 (Perturbed moments). Suppose $M(2) = a + e_2$, $M(3) = b + e_3$, and $M(6) = ab + e_6$. Then

$$\Delta = e_6 - be_2 - ae_3 - e_2e_3. \quad (5)$$

Thus a rational positive approximation with controlled signed errors can be tested against [Corollary 4.2](#) without evaluating a double sum. This is useful for machine-assisted searches, but no construction currently reaches the required D^{-2} scale.

5 The discrete moment cone and an explicit global separator

For $t \geq 0$ define the four-coordinate moment curve

$$\mathbf{v}(t) = (1, 2^t, 3^t, 6^t) \in \mathbb{R}^4,$$

and let $\mathcal{C}$ be the convex cone generated by $\mathbf{v}(n)$ for $n \in \mathbb{N}$. The target $\mathbf{v}(x)$ lies on the rank-one hypersurface $w = uv$, because $6^x = 2^x 3^x$. By [Theorem 3.1](#), every normalized point of $\mathcal{C}$ satisfies $w \geq uv$, with equality only at a single integer-exponent generator.

This nonlinear boundary statement has a remarkably simple linear dual certificate.

Theorem 5.1 (Explicit Farkas separator). *Let $k \in \mathbb{N}$ and define*

$$h_k(t) = (2^t - 2^k)(3^t - 3^{k+1}) \quad (6)$$

$$= 6^t - 3^{k+1}2^t - 2^k3^t + 2^k3^{k+1}. \quad (7)$$

Then $h_k(n) \geq 0$ for every integer $n \geq 0$, while $h_k(t) < 0$ for every $t \in (k, k+1)$.

Proof. For $n \leq k$, both factors in (6) are nonpositive; for $n \geq k+1$, both are nonnegative. If $k < t < k+1$, the first factor is positive and the second negative. $\square$

Thus the linear functional with coefficient vector

$$\lambda_k = (2^k 3^{k+1}, -3^{k+1}, -2^k, 1)$$

is nonnegative on every integer-exponent generator and strictly negative at the noninteger point $\mathbf{v}(x)$.

Corollary 5.2 (Integral separation margin). *If $a = 2^x$ and $b = 3^x$ are integers and $k < x < k+1$, then*

$$h_k(x) = -(a - 2^k)(3^{k+1} - b) \in \mathbb{Z}_{\leq -1}. \quad (8)$$

In particular, the separating margin is at least one before normalization.

Corollary 5.3 (Sup-norm distance from the positive cone). *Let $\mathbf{y}$ be any normalized positive mixture of the vectors $\mathbf{v}(n)$. If each of its four coordinates differs from the corresponding coordinate of $\mathbf{v}(x)$ by at most ε , then*

$$\varepsilon \geq \frac{1}{3^{k+1} + 2^k + 1}.$$

Proof. One has $\lambda_k \cdot \mathbf{y} \geq 0$ and $\lambda_k \cdot \mathbf{v}(x) \leq -1$. Apply the triangle inequality to $\lambda_k \cdot (\mathbf{y} - \mathbf{v}(x))$. $\square$

The same separator quantifies how much negativity a signed interpolant must contain.

Theorem 5.4 (Weighted negativity budget). *Suppose a finitely supported real sequence (c_n) satisfies*

$$\sum_n c_n \mathbf{v}(n) = \mathbf{v}(x), \quad k < x < k+1.$$

Then

$$\sum_{c_n < 0} |c_n| h_k(n) \geq (a - 2^k)(3^{k+1} - b) \geq 1. \quad (9)$$

If the support is contained in $\{0, \dots, N\}$ and $H_{k,N} = \max_{0 \leq n \leq N} h_k(n)$, then the representation forces $H_{k,N} > 0$, and the total negative mass $\eta = \sum_{c_n < 0} |c_n|$ obeys

$$\eta \geq \frac{1}{H_{k,N}}.$$

Proof. Apply the functional h_k to the representation. Its value on the target is the negative integer in (8); every term with $c_n \geq 0$ contributes nonnegatively. Moving the negative terms to the other side gives (9). The final assertion follows by bounding every $h_k(n)$ by $H_{k,N}$. $\square$

The weighted estimate is more informative than an unweighted one. Tiny negative coefficients placed at enormous exponents can carry a large separator weight, so no degree-independent lower bound for ordinary negative mass should be expected without additional growth control.

6 A cubic diagnostic interpolant

The unique cubic polynomial satisfying

$$P(1) = 1, \quad P(2) = a, \quad P(3) = b, \quad P(6) = ab$$

has the Lagrange form

$$P(z) = -\frac{(z-2)(z-3)(z-6)}{10} + a\frac{(z-1)(z-3)(z-6)}{4} - b\frac{(z-1)(z-2)(z-6)}{6} + ab\frac{(z-1)(z-2)(z-3)}{60}.$$

Writing $P(z) = c_0 + c_1z + c_2z^2 + c_3z^3$ gives

$$\begin{aligned} c_0 &= \frac{36 - 45a + 20b - ab}{10}, \\ c_1 &= \frac{-216 + 405a - 200b + 11ab}{60}, \\ c_2 &= \frac{11 - 25a + 15b - ab}{10}, \\ c_3 &= \frac{ab + 15a - 10b - 6}{60}. \end{aligned}$$

Every coefficient lies in $60^{-1}\mathbb{Z}$. By [Corollary 3.2](#), if all four coefficients were nonnegative, then P would have to be a monomial; hence $x \in \{0, 1, 2, 3\}$. In every other case at least one coefficient is at most $-1/60$.

This cubic is not intended as a proof device by itself. Its value is diagnostic: it gives the lowest-dimensional signed representation, an explicit initial point for linear programming, and a concrete instance of the negativity tax in [Theorem 5.4](#).

7 A two-moment Jensen rigidity theorem

The fourth moment $M(6)$ is not actually needed if one is willing to use the transcendental constant α explicitly.

Theorem 7.1 (Discrete Jensen rigidity). *Let (p_n) be a probability distribution on $\mathbb{N}$ with $\sum_n p_n 3^n < \infty$. Then*

$$\sum_n p_n 3^n \geq \left(\sum_n p_n 2^n \right)^\alpha. \quad (10)$$

Equality holds if and only if the distribution is a point mass.

Proof. Put $U = 2^N$. Since $3^N = U^\alpha$ and $u \mapsto u^\alpha$ is strictly convex for $\alpha > 1$, Jensen's inequality gives (10), with equality exactly when U , and hence N , is constant. $\square$

Thus a positive representation matching only the values at $1, 2, 3$ would already prove the conjecture. The advantage of the four-node covariance identity is arithmetic: it contains only integer powers and yields integral or rational defects. Jensen's inequality supplies the geometry; the covariance identity supplies the exact algebraic certificate.

For a fixed integer $A \in (2^k, 2^{k+1})$, the minimum possible value of $\mathbb{E}3^N$ among integer-valued N satisfying $\mathbb{E}2^N = A$ is obtained by a mixture of k and $k+1$. It is the chord value

$$L_k(A) = 3^k + \frac{2 \cdot 3^k}{2^k} (A - 2^k). \quad (11)$$

The strict gap $L_k(A) - A^\alpha$ is another expression of discreteness. Multiplying by 2^k produces the integer determinant developed next.

8 The integer chord cocycle

Write $x = k + \theta$ with $k = \lfloor x \rfloor$ and $0 < \theta < 1$. Define

$$\phi(\theta) = 2^{\theta+1} - 3^\theta - 1.$$

The graph $u \mapsto u^\alpha$ is strictly convex, so it lies below its chord from $(1, 1)$ to $(2, 3)$. Equivalently,

$$\phi(\theta) > 0 \quad (0 < \theta < 1), \quad (12)$$

with $\phi(0) = \phi(1) = 0$.

Theorem 8.1 (Integer chord determinant). *If $a = 2^x$ and $b = 3^x$ are integers and $x \notin \mathbb{Z}$, then*

$$E_1 := 2 \cdot 3^k (a - 2^k) - 2^k (b - 3^k) \quad (13)$$

$$= 2 \cdot 3^k a - 2^k b - 6^k \quad (14)$$

$$= 6^k \phi(\theta) \in \mathbb{Z}_{\geq 1}. \quad (15)$$

More generally, for every $n \geq 1$, put

$$m_n = \lfloor nx \rfloor, \quad \theta_n = \{nx\}.$$

Then $\theta_n \neq 0$ and

$$E_n := 2 \cdot 3^{m_n} a^n - 2^{m_n} b^n - 6^{m_n} = 6^{m_n} \phi(\theta_n) \in \mathbb{Z}_{\geq 1}. \quad (16)$$

Proof. The first identity is a rearrangement. Substituting $a = 2^k 2^\theta$ and $b = 3^k 3^\theta$ gives (15), and positivity follows from (12). For the iterated statement use $a^n = 2^{m_n} 2^{\theta_n}$ and $b^n = 3^{m_n} 3^{\theta_n}$. If $\theta_n = 0$, then $nx \in \mathbb{Z}$, so $x \in \mathbb{Q}$ and Proposition 2.1 forces $x \in \mathbb{Z}$, contrary to hypothesis. $\square$

The cocycle has exact local signatures in the primitive normalization.

Corollary 8.2 (Primitive congruence signatures). *Assume (1). For every n with $m_n \geq 2$:*

(a) *if a is odd, then $v_2(E_n) = 1$;*

(b) *if $3 \nmid b$, then $v_3(E_n) = 0$.*

Proof. In (16), if a is odd, the first term has exact 2-adic valuation 1, whereas the other two have valuation at least $m_n \geq 2$. If $3 \nmid b$, the middle term is a 3-adic unit while the first and third are divisible by 3^{m_n} . $\square$

There are also immediate endpoint exclusions. Since a^n is an integer strictly larger than 2^{m_n} and b^n is an integer strictly smaller than 3^{m_n+1} ,

$$\theta_n \geq \log_2(1 + 2^{-m_n}), \quad (17)$$

$$1 - \theta_n \geq -\log_3(1 - 3^{-m_n-1}). \quad (18)$$

These are exponentially small lower bounds. By contrast, linear-form estimates applied to $n \log a - m \log 2$ give polynomial lower bounds in n for a fixed a [4, 11]. Hence (17)–(18) are not, by themselves, competitive. The potential value of E_n lies in combining its Archimedean formula with the exact valuations in Corollary 8.2, not merely in using $E_n \geq 1$.

9 Geometric Newton interpolation on powers of two

Set

$$f(z) = z^\alpha, \quad \alpha = \log_2 3,$$

so that $f(2^j) = 3^j$. Let P_N be the Newton interpolation polynomial through the nodes $1, 2, 4, \dots, 2^N$:

$$P_N(z) = \sum_{n=0}^N c_n \prod_{j=0}^{n-1} (z - 2^j), \quad (19)$$

where an empty product is 1.

Theorem 9.1 (Closed geometric-Newton coefficients). *The coefficients in (19) are*

$$c_0 = 1, \quad c_n = \frac{\prod_{j=0}^{n-1} (3 - 2^j)}{\prod_{j=0}^{n-1} (2^n - 2^j)} \quad (n \geq 1). \quad (20)$$

For $n \geq 2$,

$$\operatorname{sgn}(c_n) = (-1)^{n-2}. \quad (21)$$

Proof. The coefficient c_n is the divided difference $[1, 2, \dots, 2^n]f$. More generally, if $q > 1$, $r > 0$, and $g(q^j) = r^j$, induction in the Newton recursion gives

$$[1, q, \dots, q^n]g = \frac{\prod_{j=0}^{n-1} (r - q^j)}{\prod_{j=0}^{n-1} (q^n - q^j)}.$$

Taking $(q, r) = (2, 3)$ gives (20). The denominator is positive. In the numerator the factors for $j = 0, 1$ are positive and all factors for $j \geq 2$ are negative, proving (21). $\square$

The sign alternation in the coefficient sequence is canceled by the sign of the geometric Newton basis beyond the interval containing an integer argument.

Theorem 9.2 (One-signed geometric tail). *Let $k \geq 1$, and let A be an integer with $2^k < A < 2^{k+1}$.*

Table 1: Exact geometric–Newton coefficients c_n for the data $2^j \mapsto 3^j$.

n	sign	$ \text{num}(c_n) $	$\text{den}(c_n)$
0	+	1	1
1	+	2	1
2	+	1	3
3	−	1	84
4	+	1	2016
5	−	13	999936
6	+	377	2015870976
7	−	22997	16384999292928
8	+	574925	106961275384233984
9	−	145456025	13992246200663952850944

Then every Newton summand

$$c_n \prod_{j=0}^{n-1} (A - 2^j), \quad n \geq k+1,$$

has the same sign, namely $(-1)^{k-1}$.

Proof. For $n = k+1$ all basis factors are positive and (21) gives sign $(-1)^{k-1}$. For $n \geq k+2$, exactly $n - k - 1$ basis factors are negative. Hence the total sign is

$$(-1)^{n-2}(-1)^{n-k-1} = (-1)^{2n-k-3} = (-1)^{k-1}.$$

□

9.1 A canonical entire interpolant

Let $(u; q)_n = \prod_{j=0}^{n-1} (1 - uq^j)$ denote the q -Pochhammer symbol. A direct factorization of (20) gives

$$c_n \prod_{j=0}^{n-1} (z - 2^j) = 2^{-n} \frac{(3; 1/2)_n (z; 1/2)_n}{(1/2; 1/2)_n}. \quad (22)$$

Consequently the series

$$H(z) = \sum_{n=0}^{\infty} 2^{-n} \frac{(3; 1/2)_n (z; 1/2)_n}{(1/2; 1/2)_n} \quad (23)$$

converges uniformly on compact subsets of $\mathbb{C}$ and defines an entire function. Indeed, the finite products in the quotient are uniformly bounded on compact z -sets, while 2^{-n} supplies geometric decay. If $z = 2^m$, then $(z; 1/2)_n = 0$ for $n > m$, so the series terminates and Newton interpolation gives

$$H(2^m) = 3^m \quad (m \in \mathbb{N}). \quad (24)$$

This construction must not be confused with a proof that $H(z) = z^\alpha$. Values on the unbounded discrete set $\{2^m\}$ do not determine an entire function without a sufficiently restrictive growth

theorem. In fact, H should be regarded as a canonical q -hypergeometric competitor to the nonentire principal-branch function z^α .

The one-signed tail gives monotone rational approximations to $H(A)$ for every integer A between consecutive powers of two. A potentially decisive theorem would compare $H(A)$ and A^α at the scale of one integer cell.

Problem 9.3 (Unit-cell comparison problem). *Establish an explicit sign and a bound of the form*

$$0 < |H(A) - A^\alpha| < \text{dist}(H(A), \mathbb{Z})$$

with the appropriate one-sided interpretation for every integer A that is not a power of two.

Such a result would immediately exclude $A^\alpha \in \mathbb{Z}$. The formulation is intentionally concrete: both $H(A)$ and its truncation error are given by rapidly convergent rational series.

9.2 The denominator wall

The unreduced denominator in (20) factors as

$$\prod_{j=0}^{n-1} (2^n - 2^j) = 2^{n(n-1)/2} \prod_{\ell=1}^n (2^\ell - 1). \quad (25)$$

Its binary logarithm is $\Theta(n^2)$. The analytic tail in (23) decays only geometrically, as $O_A(2^{-n})$. Therefore the naive plan “truncate, clear all denominators, and obtain a nonzero integer smaller than one” fails dramatically: denominator clearing loses much more than geometric convergence gains. Any arithmetic use of (23) must exploit cancellation in reduced denominators, congruences among the factors $2^\ell - 1$, or a Padé/Hermite construction with substantially better arithmetic efficiency. Basic hypergeometric methods provide a natural language for this refinement [5].

10 Signed interpolation as an optimization problem

For a degree bound N , consider the linear system

$$\sum_{n=0}^N c_n = 1, \quad \sum_{n=0}^N c_n 2^n = a, \quad \sum_{n=0}^N c_n 3^n = b, \quad \sum_{n=0}^N c_n 6^n = ab. \quad (26)$$

A natural optimization is

$$\eta_N(a, b) = \min \left\{ \sum_{n=0}^N (-c_n)_+ : (c_n) \text{ satisfies (26)} \right\}.$$

The positive-rigidity theorem says $\eta_N(a, b) > 0$ for every noninteger candidate. The dual separator gives the explicit lower bound

$$\eta_N(a, b) \geq \frac{(a - 2^k)(3^{k+1} - b)}{H_{k,N}}.$$

This linear program is attractive for two reasons. First, all data are integers and every vertex solution is rational with denominators controlled by minors of the matrix in (26). Second, dual

solutions are exponential polynomials in n , making them amenable to exact sign certification by Chebyshev-system techniques.

The main caveat is equally clear: $H_{k,N}$ grows like 6^N , so the elementary dual bound decays rapidly. A successful program needs a family of dual certificates whose coefficient norm grows much more slowly while retaining a unit integral margin at the target. This is a precise auxiliary-function problem rather than a generic appeal to linear programming.

11 Renormalized iterates and why derivative extraction is too weak

The powers $a^n = 2^{nx}$ and $b^n = 3^{nx}$ provide infinitely many rational points on the normalized curve. With $m_n = \lfloor nx \rfloor$ and $\theta_n = \{nx\}$, put

$$U_n = \frac{a^n}{2^{m_n}} = 2^{\theta_n}, \quad V_n = \frac{b^n}{3^{m_n}} = 3^{\theta_n}.$$

Whenever θ_n is small, the rational number

$$R_n = \frac{V_n - 1}{U_n - 1} = \frac{2^{m_n}(b^n - 3^{m_n})}{3^{m_n}(a^n - 2^{m_n})} \quad (27)$$

approximates α :

$$R_n = \alpha + \frac{1}{2} \log_2 3 (\log 3 - \log 2) \theta_n + O(\theta_n^2). \quad (28)$$

This follows by dividing the Taylor expansions of $e^{\theta \log 3} - 1$ and $e^{\theta \log 2} - 1$.

Dirichlet approximation gives infinitely many n with $\theta_n \ll 1/n$. But the displayed unreduced denominator and height in (27) are exponential in $m_n \asymp n$, while the error in (28) is merely $O(1/n)$. Reduced-denominator cancellation is not controlled here. Without such cancellation, the quality expressed in terms of the available height is only logarithmic, far weaker than an irrationality-measure contradiction. Higher-order Richardson combinations replace $O(\theta_n)$ by $O(\theta_n^r)$ for fixed r , but still do not overcome the visible exponential height.

This audit matters because derivative extraction from near-identity elements of the multiplicative group $\langle 2, 3 \rangle$ initially looks very promising. The calculation shows exactly why it does not compete with linear forms in logarithms: normalization creates exponentially large denominators.

12 Other routes tested and their present obstruction

12.1 Continued fractions of α

For a convergent p/q to α , both

$$3^q - 2^p \quad \text{and} \quad b^q - a^p$$

are nonzero integers when x is nonintegral. Their logarithmic discrepancies are proportional:

$$q \log b - p \log a = x(q \log 3 - p \log 2).$$

However, the integer differences themselves are of size roughly $2^p |q \log 3 - p \log 2|$ and $a^p |x(q \log 3 - p \log 2)|$. Ordinary continued-fraction accuracy is polynomial in q , while the prefactors are exponen-

tial. No “integer between zero and one” is produced.

12.2 Finite differences on 3-smooth nodes

The values s^x are integers for every $s = 2^m 3^n$. One can form divided differences and exponential-polynomial combinations on such nodes. Under a common multiplicative scaling, however, a fixed integer linear combination scales as the positive power T^x , not as the derivative-sized quantity T^{x-r} . Dividing by the Vandermonde scale destroys integrality. A successful finite-difference argument would need unusually close additive configurations of smooth numbers together with divisibility that survives normalization; the standard multiplicative grid does not provide this.

12.3 Positive approximation without exact interpolation

Bernstein polynomials and other positivity-preserving schemes approximate z^x very well on compact intervals. Yet the target vector lies strictly outside the discrete positive moment cone, with the explicit distance lower bound in [Corollary 5.3](#). Approximation alone cannot cross the separator; the arithmetic mesh must force exactness before positivity rigidity becomes available.

12.4 The canonical q -interpolant

The entire function H in (23) is arithmetically structured and rapidly computable, but equality $H(A) = A^\alpha$ is false in general and cannot be inferred from agreement on powers of two. The useful target is not equality; it is the one-integer-cell comparison in [Problem 9.3](#).

13 Computational reconnaissance

A numerical scan was performed for every integer

$$2^k < A < 2^{k+1}, \quad 1 \leq k \leq 23,$$

seeking the smallest distance from A^α to an integer in each dyadic block. A binary64 vectorized pass selected the candidate in each block; that candidate was then reevaluated with 120 decimal digits. This is exploratory evidence, not a proof: a rigorous exhaustive certificate would require directed interval arithmetic for every value or an analytic lower bound.

k	minimizing A	nearest B	$ A^\alpha - B $	$\{\log_2 A\}$
1	3	6	2.9547751e-1	5.849625e-1
2	6	17	1.1356748e-1	5.849625e-1
3	15	73	1.2410154e-1	9.068906e-1
4	24	154	2.2107357e-2	5.849625e-1
5	63	711	2.8971681e-2	9.7727992e-1
6	102	1526	4.9809728e-3	6.7242534e-1
7	163	3208	2.8016483e-4	3.4872815e-1
8	331	9859	2.6760048e-4	3.7068741e-1
9	589	24577	2.5850153e-4	2.0212382e-1
10	1537	112398	3.8544369e-4	5.8590145e-1
11	2762	284584	6.0204244e-4	4.314976e-1

k	minimizing A	nearest B	$ A^\alpha - B $	$\{\log_2 A\}$
12	5513	851059	7.0582573e-5	4.2862189e-1
13	9329	1959024	1.6530819e-5	1.8750673e-1
14	26983	10546585	3.5403609e-5	7.1976314e-1
15	33566	14906687	2.2296214e-5	3.4713003e-2
16	125362	120337724	8.289363e-6	9.3574058e-1
17	189427	231498127	4.724862e-7	5.3128245e-1
18	378854	694494381	1.4174586e-6	5.3128245e-1
19	566702	1314775264	2.1222605e-6	1.1223077e-1
20	1302553	4917240094	1.3656507e-5	3.1291065e-1
21	2774041	16296423255	3.9182644e-6	4.0355768e-1
22	4332389	33034225344	6.7526983e-5	4.6731356e-2
23	8431833	94913206833	2.6841518e-4	7.4148627e-3

The table serves two purposes. It records the relevant scale of near misses for testing auxiliary functions, and it prevents a proposed inequality from being tuned only to generic points. Any unit-cell theorem based on the canonical interpolant H should be stress-tested first at these record candidates.

14 Four focused next targets

14.1 Target A: a unit-cell theorem for the q -interpolant

Prove a one-sided comparison between $H(A)$ and A^α that is stronger than their distance to the nearest integer. The one-signed Newton tail and the explicit q -hypergeometric form make this a sharply posed analytic-arithmetic problem.

14.2 Target B: an arithmetic lower bound for weighted negativity

Strengthen (9) by showing that a rational signed interpolant of controlled height cannot place all of its negative mass at exponents where $h_k(n)$ is huge. A bound coupling coefficient denominators, support height, and separator weight could turn the unit margin into a contradiction.

14.3 Target C: an adelic obstruction for the chord cocycle

Combine the Archimedean identity $E_n = 6^{m_n} \phi(\theta_n)$ with the primitive valuations $v_2(E_n) = 1$ or $v_3(E_n) = 0$. The desired theorem would rule out an integer sequence with these fixed local signatures arising from the irrational-rotation cocycle for every n .

14.4 Target D: a machine-checked positivity core

Formalize Theorems 3.1, 5.1 and 5.4 and Proposition 4.1. These results are elementary enough to make a compact trusted kernel. A formal development would also support exact linear-program certificates produced by computation.

15 Lean formalization blueprint

The following is a proof architecture, not a claim that the code has been compiled. For finite support, define

$$F_c(t) = \sum_{i \in S} c_i t^i.$$

The key algebraic lemma is the identity

$$F_c(1)F_c(6) - F_c(2)F_c(3) = \sum_{\substack{i,j \in S \\ i < j}} c_i c_j (2^j - 2^i)(3^j - 3^i).$$

A Lean development can proceed by expanding products with `Finset.sum_mul`, splitting the double sum into diagonal and ordered pairs, and invoking positivity of natural powers.

The finite covariance and support core is formalized in `PositiveCoefficientClosure.lean`; its public endpoints are `two_mul_covariance_identity` and `finite_nonnegative_rankOne_eq_single`. The passage from finite support to the convergent infinite series remains the short truncation argument described above.

The infinite-series theorem can then be obtained either by monotone convergence or by truncating and passing to the limit. The separator theorem is even shorter: prove the sign of two monotone factors according to whether $n \leq k$ or $k + 1 \leq n$.

16 Conclusions

The work in this report does not prove the Alaoglu–Erdős conjecture. It does produce a new, compact rigidity mechanism and several exact consequences:

1. The four-node defect is a sum of strictly positive pair interactions. Zero defect forces a single integer exponent.
2. Near-zero defect quantitatively concentrates any positive representing measure, and rational weights give a discrete defect gap.
3. A four-term exponential polynomial separates every noninteger exponent from the integer moment cone with an integral margin.
4. Every signed interpolant must pay at least one unit of separator-weighted negative mass.
5. Geometric Newton coefficients for $2^j \mapsto 3^j$ have a closed product formula and produce a one-signed tail at integer arguments.
6. Every primitive hypothetical counterexample yields a positive integer cocycle with exact 2-adic or 3-adic signatures.

The positivity theorem is strong enough that the remaining issue can no longer be described merely as “find a better interpolant.” A successful proof must extract positivity, or a substitute carrying the same equality rigidity, from arithmetic data that naturally produce signed objects. Of the routes developed here, the most concrete immediate target is the unit-cell comparison for the canonical q -interpolant; the most structural is an arithmetic strengthening of the weighted-negativity budget.

A Proof of the general geometric divided-difference formula

For completeness, let $q > 1$, let r be a scalar, and let $G_n(z)$ be the Newton polynomial of degree at most n satisfying $G_n(q^m) = r^m$ for $0 \leq m \leq n$. Write

$$G_n(z) = \sum_{j=0}^n d_j \prod_{\ell=0}^{j-1} (z - q^\ell).$$

We prove

$$d_n = \frac{\prod_{\ell=0}^{n-1} (r - q^\ell)}{\prod_{\ell=0}^{n-1} (q^n - q^\ell)}.$$

For $n = 0$ this is $d_0 = 1$. Evaluating the Newton expansion at $z = q^n$ gives

$$r^n - G_{n-1}(q^n) = d_n \prod_{\ell=0}^{n-1} (q^n - q^\ell).$$

The finite q -binomial identity

$$G_{n-1}(q^n) = r^n - \prod_{\ell=0}^{n-1} (r - q^\ell)$$

follows by induction on n : both sides are polynomials of degree n in r , agree at $r = q^j$ for $0 \leq j < n$, and have the same leading coefficient. Substitution gives the claimed formula.

B Reproducible numerical scan

The following compact script reproduces the dyadic-block scan in [Section 13](#). It is intentionally labeled experimental rather than rigorous.

```
import numpy as np
import mpmath as mp

mp.mp.dps = 120
alpha_mp = mp.log(3) / mp.log(2)
alpha = float(alpha_mp)

for k in range(1, 24):
    lo, hi = 2**k + 1, 2**(k+1)
    best_A, best_d = None, 1.0
    for s in range(lo, hi, 1_000_000):
        e = min(s + 1_000_000, hi)
        A = np.arange(s, e, dtype=np.float64)
        Y = np.exp(alpha * np.log(A))
        D = np.abs(Y - np rint(Y))
        j = int(np.argmax(D))
        if float(D[j]) < best_d:
            best_d = float(D[j])
            best_A = s + j
    y = mp.power(mp.mpf(best_A), alpha_mp)
    nearest = mp.nint(y)
```

```
print(k, best_A, nearest, abs(y-nearest))
```

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